

ZimaOS Monitor for Home Assistant

Complete Installation Guide



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Prerequisites

Before installing the ZimaOS Monitor integration, ensure you have:

System Requirements

- **Home Assistant:** Version 2023.1.0 or newer
- **ZimaOS:** Any version with SSH capability
- **Network:** Both systems must be on the same network or accessible via network routing
- **Permissions:** Administrator access to both Home Assistant and ZimaOS

Required Information

Gather the following information before starting:

- ZimaOS IP address (e.g., 192.168.1.100)
 - SSH username (typically zimaos)
 - SSH password
 - SSH port (default: 22)
-

Preparing Your ZimaOS System

Step 1: Enable SSH Access on ZimaOS

1. Access ZimaOS Web Interface

- Open your web browser
- Navigate to your ZimaOS IP address (e.g., http://192.168.1.100)
- Log in with your administrator credentials

2. Navigate to SSH Settings

- Click on **Settings** in the main menu
- Select **System** from the submenu
- Click on **SSH or Terminal Access**

3. Enable SSH Service

- Toggle the SSH service to **ON**
- Note the SSH port (default is 22)
- If prompted, set or confirm your SSH password
- Click **Save** or **Apply**

4. Verify SSH Access (Optional but recommended) From any computer with SSH client:

```
bash  
  
ssh zimaos@YOUR_ZIMAOS_IP
```

If successful, you'll be prompted for a password. Enter it to confirm access works.

Step 2: Check User Permissions

Ensure your SSH user has the necessary permissions:

```
bash

# After logging in via SSH, test these commands:
df -h      # Should show disk information
free -m    # Should show memory information
top -bn1   # Should show process information
```

If any command fails, you may need to adjust user permissions or use a different user account.

Installation Methods

You can install the ZimaOS Monitor integration using one of three methods:

Method 1: HACS Installation (Recommended)

1.1 Prerequisites for HACS

- HACS must be installed in your Home Assistant instance
- If not installed, visit hacs.xyz for installation instructions

1.2 Add Custom Repository

1. Open HACS in Home Assistant

- Navigate to **HACS** in the sidebar
- Click on **Integrations**

2. Add Custom Repository

- Click the three-dot menu (⋮) in the top right
- Select **Custom repositories**
- In the dialog that appears:
 - **Repository:**
 - **Category:** Select
- Click **ADD**

3. Install the Integration

- Search for "ZimaOS Monitor" in HACS
- Click on the integration when it appears
- Click **INSTALL**
- Select the latest version

- Click **INSTALL** again to confirm

4. Restart Home Assistant

- Go to **Settings** → **System** → **Restart**
- Click **Restart Home Assistant**
- Wait for Home Assistant to fully restart (usually 1-2 minutes)

Method 2: Manual Installation

2.1 Download the Integration

1. Download from GitHub

- Visit: <https://github.com/chicoxaager/zimaos-hass-integration/releases>
- Download the latest release (ZIP file)
- Extract the ZIP file to your computer

2. Prepare the Files

- Inside the extracted folder, locate the `zimaos` directory
- This directory contains all necessary files:

```
zimaos/  
├── __init__.py  
├── config_flow.py  
├── const.py  
├── hacs.json  
├── icon.png  
├── manifest.json  
├── sensor.py  
├── strings.json  
└── zimaos_api.py
```

2.2 Copy Files to Home Assistant

Option A: Using File Editor Add-on

1. Install File Editor add-on from Supervisor → Add-on Store
2. Open File Editor
3. Navigate to `/config/custom_components/`
4. Create a new folder named `zimaos`
5. Upload all files from the downloaded `zimaos` folder

Option B: Using Samba Share

1. Enable Samba Share add-on in Home Assistant
2. Access your Home Assistant config folder from your computer
3. Navigate to `custom_components` folder (create if doesn't exist)
4. Copy the entire `zimaos` folder here

Option C: Using SSH/Terminal

```
bash

# Connect to Home Assistant via SSH
ssh YOUR_HA_USER@YOUR_HA_IP

# Navigate to custom_components
cd /config/custom_components

# Create zimaos directory
mkdir zimaos

# Copy files (example using wget)
cd zimaos
wget https://raw.githubusercontent.com/chicoahaager/zimaos-hass-integration/main/zimaos/__init__.py
wget https://raw.githubusercontent.com/chicoahaager/zimaos-hass-integration/main/zimaos/manifest.json
# ... repeat for all files
```

2.3 Verify File Structure

Ensure your file structure looks like this:

```
/config/
├── custom_components/
│   └── zimaos/
│       ├── __init__.py
│       ├── config_flow.py
│       ├── const.py
│       ├── hacs.json
│       ├── icon.png
│       ├── manifest.json
│       ├── sensor.py
│       ├── strings.json
│       └── zimaos_api.py
```

2.4 Restart Home Assistant

Required after manual installation!

1. Go to **Settings** → **System**
2. Click **Restart**
3. Wait for complete restart

Method 3: Git Clone (For Developers)

```
bash

# SSH into Home Assistant
cd /config/custom_components
git clone https://github.com/chicoxaager/zimaos-hass-integration.git zimaos-temp
mv zimaos-temp/zimaos ./
rm -rf zimaos-temp
```

Configuration

Step 1: Add the Integration

1. **Navigate to Integrations**
 - Go to **Settings** → **Devices & Services**
 - Click + **ADD INTEGRATION** button
2. **Search for ZimaOS**
 - Type "ZimaOS" in the search box
 - Click on "ZimaOS Monitor" when it appears
 - If it doesn't appear, try:
 - Clear browser cache (Ctrl+F5)
 - Restart Home Assistant again
 - Check installation files are correct

Step 2: Configure Connection

When the configuration dialog appears:

1. **Enter Connection Details**
 - **Host IP Address:** Your ZimaOS IP (e.g.,)
 - **SSH Username:** Usually (or your custom username)
 - **SSH Password:** Your SSH password
 - **SSH Port:** Default is (change if you use a custom port)
2. **Submit Configuration**
 - Click **SUBMIT**

- Wait for connection test (may take 10-30 seconds)

3. Handle Potential Errors

- **"Authentication Failed"**: Check username/password
- **"Connection Failed"**: Verify IP address and SSH is enabled
- **"Timeout"**: Check firewall settings or network connectivity

Step 3: Successful Configuration

Once configured successfully:

- You'll see "Success!" message
 - The integration will appear in your integrations list
 - A new device "ZimaOS (YOUR_IP)" will be created
-

Verification & Testing

Step 1: Check Device Creation

1. Go to **Settings** → **Devices & Services**
2. Click on **ZimaOS Monitor** integration
3. You should see:
 - 1 Device
 - 30+ Entities (sensors)

Step 2: View Available Sensors

Click on the device to see all sensors. You should have:

System Sensors:

- Health Score (with A-F grading)
- CPU Usage
- Memory Usage/Used/Total/Available
- CPU Temperature
- Uptime

Storage Sensors:

- Disk Usage Root
- Disk Usage DATA (if DATA partition exists)
- Disk I/O Read/Write speeds

Network Sensors:

- Network RX/TX Total
- Network Connections
- SSH Connections

Performance Sensors:

- Load Average (1/5/15 min)
- Process Count
- CPU Cores
- Swap Usage

Docker Sensors (if Docker installed):

- Docker Containers Running
- Docker Containers Total
- Docker Images Count

Step 3: Test Sensor Updates

1. Navigate to any sensor
2. Click on it to see the graph
3. Verify it updates every 30 seconds
4. Check that values are realistic

Step 4: Create a Test Dashboard

yaml

Add this to a dashboard to test

type: vertical-stack

cards:

- **type:** entity

entity: sensor.zimaos_health_score

name: System Health

- **type:** horizontal-stack

cards:

- **type:** gauge

entity: sensor.zimaos_cpu_usage

max: 100

severity:

green: 0

yellow: 60

red: 80

- **type:** gauge

entity: sensor.zimaos_memory_usage

max: 100

severity:

green: 0

yellow: 70

red: 90

- **type:** gauge

entity: sensor.zimaos_disk_usage_root

max: 100

severity:

green: 0

yellow: 70

red: 85

Troubleshooting

Common Issues and Solutions

1. Integration Not Appearing After Installation

Symptoms: Can't find "ZimaOS" when adding integration

Solutions:

- Clear browser cache: `Ctrl+Shift+F5`
- Check files are in correct location: `/config/custom_components/zimaos/`
- Verify all 9 files are present
- Check Home Assistant logs: **Settings** → **System** → **Logs**

- Restart Home Assistant completely

2. Connection Failed Error

Symptoms: Error when configuring the integration

Solutions:

```
bash

# Test from Home Assistant host
ping YOUR_ZIMAOS_IP

# Test SSH connection
ssh zimaos@YOUR_ZIMAOS_IP

# Check firewall on ZimaOS
sudo ufw status

# If firewall blocks SSH:
sudo ufw allow 22/tcp
```

3. Authentication Failed

Solutions:

- Verify username (case-sensitive)
- Reset SSH password on ZimaOS
- Try with a different user account
- Check SSH allows password authentication:

```
bash

# On ZimaOS
sudo nano /etc/ssh/sshd_config
# Ensure: PasswordAuthentication yes
sudo systemctl restart sshd
```

4. Missing Sensors

Some sensors only appear when available:

- Docker sensors: Only if Docker is installed
- DATA partition: Only if /DATA mount exists
- Temperature: Only if thermal zones are available

To check availability:

```
bash

# On ZimaOS via SSH
docker --version    # For Docker
df -h | grep DATA  # For DATA partition
cat /sys/class/thermal/thermal_zone*/temp # For temperature
```

5. Sensors Not Updating

Solutions:

- Check integration status in Devices & Services
- Look for errors in logs
- Manually reload integration:
 - Click three dots on integration
 - Select "Reload"
- Check SSH connection is stable

Checking Logs

1. Home Assistant Logs

Settings → System → Logs

Look for entries containing "zimaos"

2. Enable Debug Logging Add to `configuration.yaml`:

```
yaml

logger:
  default: info
logs:
  custom_components.zimaos: debug
```

3. SSH Connection Test

```
bash

# From Home Assistant SSH
ssh -v zimaos@YOUR_ZIMAOS_IP
```

Advanced Configuration

Customizing Update Interval

Default update interval is 30 seconds. To change:

1. Edit `/config/custom_components/zimaos/const.py`
2. Find line: `SCAN_INTERVAL = 30`
3. Change to desired seconds (e.g., `60` for 1 minute)
4. Restart Home Assistant

Creating Automations

Example: Alert on High CPU

```
yaml

automation:
  - alias: "ZimaOS High CPU Alert"
    trigger:
      - platform: numeric_state
        entity_id: sensor.zimaos_cpu_usage
        above: 90
        for: "00:05:00"
    action:
      - service: notify.mobile_app_your_phone
        data:
          title: "ZimaOS Alert"
          message: "CPU usage above 90% for 5 minutes!"
```

Example: Daily Health Report

```
yaml
```

automation:

- alias: "ZimaOS Daily Health Report"

trigger:

- platform: time
- at: "09:00:00"

action:

- service: notify.mobile_app_your_phone

data:

title: "ZimaOS Health Report"

message: >

Health Score: {{ states('sensor.zimaos_health_score') }}%

(Grade: {{ state_attr('sensor.zimaos_health_score', 'grade') }})

CPU: {{ states('sensor.zimaos_cpu_usage') }}%

Memory: {{ states('sensor.zimaos_memory_usage') }}%

Uptime: {{ states('sensor.zimaos_uptime') }} days

Custom Lovelace Cards

Health Score with Grade:

yaml

type: custom:mushroom-entity-card

entity: sensor.zimaos_health_score

name: ZimaOS Health

icon_color: >

{% set score = states('sensor.zimaos_health_score')|int %}

{% if score >= 90 %} green

{% elif score >= 70 %} yellow

{% else %} red

{% endif %}

secondary_info: >

Grade: {{ state_attr('sensor.zimaos_health_score', 'grade') }} -

{{ state_attr('sensor.zimaos_health_score', 'status') }}

Support & Resources

Getting Help

1. **GitHub Issues:** [Report bugs or request features](#)
2. **GitHub Discussions:** [Community support](#)
3. **Home Assistant Community:** Search for "ZimaOS" in forums

Useful Links

- **Project Repository:** <https://github.com/chicohaager/zimaos-hass-integration>
- **Latest Releases:** <https://github.com/chicohaager/zimaos-hass-integration/releases>
- **HACS Documentation:** <https://hacs.xyz>
- **Home Assistant:** <https://www.home-assistant.io>

Providing Feedback

When reporting issues, please include:

1. Home Assistant version
 2. ZimaOS Monitor integration version
 3. Error messages from logs
 4. Steps to reproduce the issue
 5. Your system configuration (sanitized)
-

Final Notes

- **Updates:** The integration will update automatically if installed via HACS
- **Security:** SSH credentials are stored encrypted in Home Assistant
- **Performance:** The integration uses minimal resources (~1-2% CPU during updates)
- **Compatibility:** Works with all ZimaOS versions that support SSH

Congratulations! Your ZimaOS Monitor integration should now be fully operational. Enjoy monitoring your ZimaOS system directly from Home Assistant!